

# Minimizing Transfer Function Variance in Sallen-Key Cascades

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## Abstract

Sallen-Key active filter architectures are ubiquitous in modern integrated circuit design. In this paper, we use Schoeffler's variance formula to derive a strict, closed-form bound on the variance of the frequency response's magnitude as it relates to component ratios, quality factor, and component tolerances. We then use this bound to prove the most variation-resilient Sallen-Key architecture has unity gain and equal resistor values for the lowpass case and equal capacitor values for the highpass case. We also bound second-order effects on transfer function variance in terms of component sensitivities. We then make design recommendations for variation resilient Sallen-Key architectures with common E24 component values and run Monte Carlo simulations to loosely verify results. We suspect this bound to be useful in Electronic Design Applications (EDA) and in giving insights to designers picking tolerances for Sallen-Key stages.

## I. INTRODUCTION

Since their introduction, active Sallen-Key filters are often used in integrated circuits due to their simplicity (low component count) and ease of miniaturization (unlike inductors) [1]. However, a high quality factors, we find that designers are typically recommended to use other, less sensitive architectures like Multiple Feedback (MFB) architectures [2]. However, it's still of interest to know a bound for Sallen-Key variance to make an informed choice to switch to MFB architectures. We take the standard approach of using sensitivity analysis and relating it to variance as first described by Schoeffler [3]. Though many works look at sensitivity for circuits like Sallen-Keys as in [4], a closed-form, strict bound is still useful in understanding how variance relates to tolerances and quality factor. By relating tolerances to a computable bound on variance, we can tune our bound to some statistical certainty, which may be helpful for analysis with the intent to mass-produce the filter. Though this analysis, we mathematically prove that the  $K = 1$ , equal-resistor-value case is optimal in minimizing sensitivity in a Sallen-Key lowpass, and the  $K = 1$ , equal-capacitor-value case is optimal, as shown by [5]. We also bound second-order effects and simulate our results in LTSpice to ensure accuracy.

## II. SENSITIVITY AND VARIANCE DEFINITIONS

We will use normalized sensitivity and variance to discuss parameter variations in filters sensibly, which are defined below:

**Definition 1.** *The **normalized sensitivity** of variable  $y$  with respect to variable  $x$  is the following:*

$$S_x^y := \frac{\partial(\ln y)}{\partial(\ln x)} = x \frac{\partial(\ln y)}{\partial x} = \frac{x}{y} \left( \frac{\partial y}{\partial x} \right) \quad (1)$$

**Definition 2.** *The **normalized log variance** of variable  $x$  is  $\tau_x^2 := \left( \frac{\sigma_x}{\mathbb{E}(x)} \right)^2 = \text{Var}\left(\frac{\Delta x}{x}\right)$*

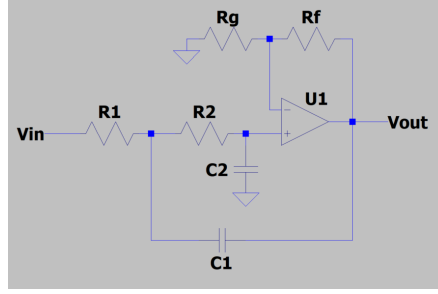


Fig. 1: A Standard Sallen-Key Architecture

### III. SALLLEN-KEY LOWPASS CASCADES

Consider a general  $n$ -th order filter made of a cascade of Sallen-Keys, where  $n$  is even. As each block's op-amp acts as a buffer preventing loading effects, we know the total transfer function is simply the product of all the blocks, like so:

$$H(s) = \prod_{i=1}^{n/2} H_i(s) \quad (2)$$

where  $H_i(s)$  dictates the transfer function of a standard Sallen-Key lowpass, such as in Figure 1. It's a standard result that this topology behaves like a second-order system,<sup>1</sup> like so:

$$H_i(s) = \frac{K}{1 + \frac{s}{\omega_i Q_i} + \frac{s^2}{\omega_i^2}} \quad (3)$$

$$\begin{cases} \omega_i = (\sqrt{R_{1,i} R_{2,i} C_{1,i} C_{2,i}})^{-1} \\ Q_i = \frac{\omega_i^{-1}}{C_{2,i}(R_{1,i} + R_{2,i}) + (1-K)C_{1,i}R_{1,i}} \\ K = 1 + \frac{R_{f,i}}{R_{g,i}} \end{cases} \quad (4)$$

If we assume we can construct  $K$  from resistor ratios accurately enough that we can ignore sensitivity in  $K$ , we can quite easily compute the sensitivity of parameters with respect to our resistors and capacitors.

$$\begin{cases} S_{R_{1,i}}^{\omega_i} = S_{R_{2,i}}^{\omega_i} = S_{C_{1,i}}^{\omega_i} = S_{C_{2,i}}^{\omega_i} = -\frac{1}{2} \\ S_{R_{1,i}}^{Q_i} = -\frac{1}{2} + \sqrt{\frac{R_{2,i}C_{2,i}}{R_{1,i}C_{1,i}}} Q_i \\ S_{R_{2,i}}^{Q_i} = \frac{1}{2} - \sqrt{\frac{R_{2,i}C_{2,i}}{R_{1,i}C_{1,i}}} Q_i \\ S_{C_{1,i}}^{Q_i} = \frac{1}{2} + (K-1)\sqrt{\frac{R_{1,i}C_{1,i}}{R_{2,i}C_{2,i}}} Q_i \\ S_{C_{2,i}}^{Q_i} = -\frac{1}{2} - (K-1)\sqrt{\frac{R_{1,i}C_{1,i}}{R_{2,i}C_{2,i}}} Q_i \end{cases} \quad (5)$$

It's important to note that  $S_{R_{1,i}}^{Q_i} = -S_{R_{2,i}}^{Q_i}$  and  $S_{C_{1,i}}^{Q_i} = -S_{C_{2,i}}^{Q_i}$ , which is consistent with Euler's homogeneous function theorem. To elaborate, since  $Q_i$  is unitless, the sum of the  $Q_i$ -sensitivities is zero. We'll introduce a definition to make the math easier to parse through.

$$\rho_i^2 := \frac{R_{2,i}C_{2,i}}{R_{1,i}C_{1,i}} \quad (6)$$

We need to find the sensitivity of  $|H_i(j\omega)|$  with respect to  $\omega_i$  and  $Q_i$ . We then use the multivariate chain rule.

$$S_{x_i}^{|H_i(j\omega)|} = S_{\omega_i}^{|H_i(j\omega)|} (S_{x_i}^{\omega_i}) + S_{Q_i}^{|H_i(j\omega)|} (S_{x_i}^{Q_i}) \quad (7)$$

<sup>1</sup>Assuming an ideal op-amp. That is, we're working well within the op-amp's bandwidth.

where we assume  $K$  sensitivity is negligible and  $x_i \in \{R_{1,i}, R_{2,i}, C_{1,i}, C_{2,i}\} =: \mathcal{P}_i$ . We can get a full variance of each gain stage from Schoeffler variance.

$$\tau_{|H_i(j\omega)|}^2 \approx \sum_{x_i \in \mathcal{P}_i} \left( S_{x_i}^{|H_i(j\omega)|} \right)^2 \tau_{x_i}^2 \quad (8)$$

Our goal now is to strictly bound the variance over all  $\omega > 0$ , and then minimize the bound. We know from standard results considering a second order system (as in [4]):

$$\begin{cases} S_{\omega_i}^{|H_i(j\omega)|}(\Omega) = \Omega^2 \left( \frac{2(\Omega^2-1) + \frac{1}{Q_i^2}}{(1-\Omega^2)^2 + \frac{\Omega^2}{Q_i^2}} \right) \\ S_{Q_i}^{|H_i(j\omega)|}(\Omega) = \frac{\frac{\Omega^2}{Q_i^2}}{(1-\Omega^2)^2 + \frac{\Omega^2}{Q_i^2}} \end{cases} \quad (9)$$

where  $\Omega := \frac{\omega}{\omega_i}$  is the normalized frequency. We'll start by bounding the sensitivity for any given stage. Noting that  $(1 - \Omega^2)^2 \geq 0$ ,

$$S_{Q_i}^{|H_i(j\omega)|}(\Omega) \leq \frac{\frac{\Omega^2}{Q_i^2}}{\frac{\Omega^2}{Q_i^2}} = 1 = S_{Q_i}^{|H_i(j\omega)|}(1) \quad (10)$$

Thus, we have the strictest upper bound possible for our  $Q$  factor sensitivity. Bounding the resonant frequency sensitivity is a little more intensive, but if we define  $x := \Omega^2$  and  $k := 2 - \frac{1}{Q_i^2}$ ,

$$S_{\omega_i}^{|H_i(j\omega)|}(\Omega) = \Omega^2 \left( \frac{2(\Omega^2 - 1) + \frac{1}{Q_i^2}}{(1 - \Omega^2)^2 + \frac{\Omega^2}{Q_i^2}} \right) = \frac{2x - k}{x - k + \frac{1}{x}} \implies \quad (11)$$

$$\frac{\partial S_{\omega_i}^{|H_i(j\omega)|}}{\partial \omega} = \frac{-k + \frac{2}{x} + \frac{2x-k}{x^2}}{(x - k + \frac{1}{x})^2} \left( \frac{\partial x}{\partial \omega} \right) \quad (12)$$

We achieve a maximum sensitivity when  $\frac{\partial S_{\omega_i}^{|H_i(j\omega)|}}{\partial \omega} = 0$ , so since  $x > 0 \implies \frac{\partial x}{\partial \omega}, (x - k - \frac{1}{x})^2 > 0$ :

$$x = \frac{2 \pm \sqrt{4 - k^2}}{k} \quad (13)$$

It's easy to show that the positive root is the local maximum. Substituting back into our sensitivity equation, we see that:

$$\max(S_{\omega_i}^{|H_i(j\omega)|}) = 1 + \frac{Q_i}{\sqrt{1 - \frac{1}{4Q_i^2}}} =: S_{\max} \quad (14)$$

Thus, we can bound the stage variance.

$$\tau_{|H_i(j\omega)|}^2 = \sum_{x_i \in \mathcal{P}_i} \left( S_{\omega_i}^{|H_i(j\omega)|}(S_{x_i}^{\omega_i}) + S_{Q_i}^{|H_i(j\omega)|}(S_{x_i}^{Q_i}) \right)^2 \tau_{x_i}^2 \quad (15)$$

$$\leq \sum_{x_i \in \mathcal{P}_i} \left( -\frac{1}{2} S_{\max} + S_{x_i}^{Q_i} \right)^2 \tau_{x_i}^2 \quad (16)$$

$$= \sum_{x_i \in \mathcal{P}_i} \tau_{x_i}^2 \left( \frac{1}{4} S_{\max}^2 - S_{\max} S_{x_i}^{Q_i} + (S_{x_i}^{Q_i})^2 \right) \quad (17)$$

Let's assume for any given stage that there's a uniform resistor tolerance  $\tau_{R,i}^2$  and capacitor tolerance  $\tau_{C,i}^2$ . Then, we can express our tolerance bound in terms of  $\tau_{R,i}^2$  and  $\tau_{C,i}^2$ .

$$\tau_{|H_i(j\omega)|}^2 \leq \tau_{R,i}^2 \left( \frac{1}{2} S_{\max}^2 + \sum_{x_i \in \{R_{1,i}, R_{2,i}\}} \left( (S_{x_i}^{Q_i})^2 - S_{\max} S_{x_i}^{Q_i} \right) \right) \quad (18)$$

$$+ \tau_{C,i}^2 \left( \frac{1}{2} S_{\max}^2 + \sum_{x_i \in \{C_{1,i}, C_{2,i}\}} \left( (S_{x_i}^{Q_i})^2 - S_{\max} S_{x_i}^{Q_i} \right) \right) \quad (19)$$

With our previously stated sensitivities, the cross-terms cancel.

$$\sum_{x_i \in \{R_{1,i}, R_{2,i}\}} -S_{\max} S_{x_i}^{Q_i} = 0 \quad \sum_{x_i \in \{C_{1,i}, C_{2,i}\}} -S_{\max} S_{x_i}^{Q_i} = 0 \quad (20)$$

Our final stage tolerance can be expressed as a polynomial in  $\rho_i$ .

$$\tau_{|H_i(j\omega)|}^2 \leq \frac{\tau_{R,i}^2 + \tau_{C,i}^2}{2} (S_{\max}^2) + 2\tau_{R,i}^2 (S_{R_{1,i}}^{Q_i})^2 + 2\tau_{C,i}^2 (S_{C_{1,i}}^{Q_i})^2 \quad (21)$$

$$= \frac{\tau_{R,i}^2 + \tau_{C,i}^2}{2} (S_{\max}^2 + 1) + 2\tau_{C,i}^2 (K-1) \rho_i^{-1} Q_i - 2\tau_{R,i}^2 \rho_i Q_i + 2\tau_{C,i}^2 (K-1)^2 \rho_i^{-2} Q_i^2 + 2\tau_{R,i}^2 \rho_i^2 Q_i^2 \quad (22)$$

Now suppose we want to optimize our component ratio  $\rho_i > 0$ , so we can minimize our bound when  $\frac{\partial(\text{bound})}{\partial \rho_i} = 0$ .

$$\frac{\partial(\text{bound})}{\partial \rho_i} = -2\tau_{C,i}^2 (K-1) \rho_i^{-2} Q_i - 2\tau_{R,i}^2 Q_i - 4\tau_{C,i}^2 (K-1)^2 \rho_i^{-3} Q_i^2 + 4\tau_{R,i}^2 \rho_i Q_i^2 = 0 \implies \quad (23)$$

$$2\tau_{R,i}^2 Q_i^2 \rho_i^4 - \tau_{R,i}^2 Q_i \rho_i^3 - \tau_{C,i}^2 (K-1) \rho_i Q_i - 2\tau_{C,i}^2 (K-1)^2 Q_i^2 = 0 \quad (24)$$

To use this to find an optimal  $\rho_i$  for  $K \neq 1$ , it's best solved numerically, picking the minimum of the positive real roots. However, if  $K = 1$  (as is common and typically optimal in many Sallen-Key stages), the capacitor terms vanish.

$$2\tau_{R,i}^2 Q_i^2 \rho_i^4 = \tau_{R,i}^2 Q_i \rho_i^3 \implies \rho_i = \frac{1}{2Q_i} \quad (25)$$

To then use this variance bound over all stages, we use variance properties to derive  $\tau_{|H(j\omega)|}^2$  in terms of  $\tau_{|H_i(j\omega)|}^2$  using Eq. (2) and then substituting our bounds for each stage. We'll proceed by noting the following.

$$\text{Var}(\ln(x)) \approx (d(\ln(x)))^2 = \frac{dx}{x} \approx \tau_x^2 \quad (26)$$

From Eq. (2), we get that

$$\ln(|H(j\omega)|) = \sum_{i=1}^{n/2} \ln(|H_i(j\omega)|) \implies \quad (27)$$

$$\text{Var}(\ln(|H(j\omega)|)) = \sum_{i=1}^{n/2} \text{Var}(\ln(|H_i(j\omega)|)) \implies (\approx) \quad (28)$$

$$\tau_{|H(j\omega)|}^2 = \sum_{i=1}^{n/2} \tau_{|H_i(j\omega)|}^2 \quad (29)$$

#### IV. SALLÉN-KEY HIGHPASS TRANSFORMATION

We can use all the techniques used in the previous section on a highpass filter, which we make from a lowpass filter where  $s \rightarrow \frac{\omega_i^2}{s}$ . Then, we get the transfer function of a highpass stage.

$$H_i(s) = H_{LP,i} \left( \frac{\omega_i^2}{s} \right) = \frac{K \left( \frac{s^2}{\omega_i^2} \right)}{\left( \frac{s^2}{\omega_i^2} \right) + \frac{s}{\omega_i Q_i} + 1} \quad (30)$$

where  $\omega_i, Q_i$  similar as before, except with the resistor and capacitor sensitivities flipped in our quality factor expression (confirmed in **adi minimizing sensitivity**).

$$\begin{cases} S_{R_{1,i}}^{\omega_i} = S_{R_{2,i}}^{\omega_i} = S_{C_{1,i}}^{\omega_i} = S_{C_{2,i}}^{\omega_i} = -\frac{1}{2} \\ S_{C_{1,i}}^{Q_i} = -\frac{1}{2} + \sqrt{\frac{R_{2,i}C_{2,i}}{R_{1,i}C_{1,i}}} Q_i \\ S_{C_{2,i}}^{Q_i} = \frac{1}{2} - \sqrt{\frac{R_{2,i}C_{2,i}}{R_{1,i}C_{1,i}}} Q_i \\ S_{R_{1,i}}^{Q_i} = \frac{1}{2} + (K-1) \sqrt{\frac{R_{1,i}C_{1,i}}{R_{2,i}C_{2,i}}} Q_i \\ S_{R_{2,i}}^{Q_i} = -\frac{1}{2} - (K-1) \sqrt{\frac{R_{1,i}C_{1,i}}{R_{2,i}C_{2,i}}} Q_i \end{cases} \quad (31)$$

Now, we need to compute  $S_{Q_i}^{|H_i(j\omega)|}, S_{\omega_i}^{|H_i(j\omega)|}$ . Defining  $\Omega$  as before, we get that,

$$S_{x_i}^{|H_i(j\omega)|} = \frac{x_i}{2} \left( \frac{\partial(\ln(|H_i(j\omega)|^2))}{\partial x_i} \right) = \left( \frac{x_i}{2} \right) \frac{\partial}{\partial x_i} \left[ \ln(K^2\Omega^4) - \ln \left( (1-\Omega^2)^2 + \frac{\Omega^2}{Q_i^2} \right) \right] \quad (32)$$

Since  $\frac{\partial}{\partial Q_i}(\ln(K^2\Omega^4)) = 0$ ,  $S_{Q_i}^{|H_i(j\omega)|}$  remains unchanged from our previous part. However,  $S_{\omega_i}^{|H_i(j\omega)|}$  does change.

$$S_{\omega_i}^{|H_i(j\omega)|} = \left( \frac{\omega_i}{2} \right) \frac{\partial}{\partial \omega_i} [\ln(K^2\Omega^4)] + \Omega^2 \left( \frac{2(\Omega^2 - 1) + \frac{1}{Q_i^2}}{(1-\Omega^2)^2 + \frac{\Omega^2}{Q_i^2}} \right) \quad (33)$$

We then simplify the first term.

$$\left( \frac{\omega_i}{2} \right) \frac{\partial}{\partial \omega_i} [\ln(K^2\Omega^4)] = -2 \quad (34)$$

Thus, from our previous assertion in Eq. (14), we can reassert for the highpass filter that

$$\max(|S_{\omega_i}^{|H_i(j\omega)|}|) = \max(\max(S_{\omega_i}^{|H_i(j\omega)|}), |\min(S_{\omega_i}^{|H_i(j\omega)|})|) = \left| -1 - \frac{Q_i}{\sqrt{1 - \frac{1}{4Q_i^2}}} \right| \quad (35)$$

$$= 1 + \frac{Q_i}{\sqrt{1 - \frac{1}{4Q_i^2}}} =: S_{\max} \quad \max(S_{Q_i}^{|H_i(j\omega)|}) = 1 \quad (36)$$

We find the maximum absolute value of the corner frequency sensitivity as we square  $S_{\max}$  in Eq. (21). We then can continue our reasoning as before from Eq. (21) onward, where:

$$\tau_{|H_i(j\omega)|}^2 \leq \frac{\tau_{R,i}^2 + \tau_{C,i}^2}{2} (S_{\max}^2 + 1) + 2\tau_{R,i}^2 (K-1) \rho_i^{-1} Q_i - 2\tau_{C,i}^2 \rho_i Q_i + 2\tau_{R,i}^2 (K-1)^2 \rho_i^{-2} Q_i^2 + 2\tau_{C,i}^2 \rho_i^2 Q_i^2 \quad (37)$$

We still get the same optimal  $\rho_i$  in this case. We can also easily show this implies equal capacitance by [6].

## V. SALLER-KEY WIDEBAND BANDPASS FILTER

A common construction of a  $n$ -th order bandpass filter is an  $n$ -th order lowpass followed by an  $n$ -th order highpass. We can express our tolerance as their sum.

$$\tau_{|H(j\omega)|}^2 = \sum_{i=1}^{n/2} \left( \tau_{|H_i(j\omega)|,LP}^2 + \tau_{|H_i(j\omega)|,HP}^2 \right) = \sum_{i=1}^{n/2} \tau_{|H_i(j\omega)|}^2 \quad (38)$$

where

$$\tau_{|H_i(j\omega)|}^2 \leq \tau_{RC,i}^2 (S_{\max}^2 + 1) + 2\tau_{RC,i}^2 (K-1) \rho_i^{-1} Q_i - 2\tau_{RC,i}^2 \rho_i Q_i + 2\tau_{RC,i}^2 (K-1)^2 \rho_i^{-2} Q_i^2 + 2\tau_{RC,i}^2 \rho_i^2 Q_i^2 \quad (39)$$

and  $\tau_{RC,i}^2 = \tau_{R,i}^2 + \tau_{C,i}^2$

## VI. BOUND ON SECOND-ORDER EFFECTS WITH SCHOEFLER'S VARIANCE FORMULA

Recall that Schoeffler's variance formula (as stated in Eq. (8)) is a first-order approximation of variance. Now suppose we define  $M_i(\omega, \mathcal{P}_i) := |H_i(j\omega)|$  to be the magnitude of the transfer function as a function of  $\mathcal{P}_i$ . We can then expand this via a Taylor series.

$$\frac{\Delta M_i}{M_i} = \sum_{x \in \mathcal{P}_i} S_x^{M_i} \frac{\Delta x}{x} + \frac{1}{2} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} \frac{xy}{M_i} \frac{\partial^2 M_i}{\partial x \partial y} \left( \frac{\Delta x}{x} \right) \left( \frac{\Delta y}{y} \right) + \mathcal{O}(\Delta x^3) \quad (40)$$

Now we define a second-order sensitivity element.

$$S_{xy}^{M_i} := \frac{xy}{M_i} \left( \frac{\partial^2 M_i}{\partial x \partial y} \right) \quad (41)$$

Then, by taking the variance of both sides,

$$\tau_{M_i}^2 = \text{Var} \left( \frac{\Delta M_i}{M_i} \right) = \sum_{x \in \mathcal{P}_i} (S_x^{M_i})^2 \tau_x^2 + \text{Var} \left( \frac{1}{2} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} S_{xy}^{M_i} \left( \frac{\Delta x}{x} \right) \left( \frac{\Delta y}{y} \right) \right) + \text{Var}(\mathcal{O}(\Delta x^3)) \quad (42)$$

Expanding our second term, we get that

$$\text{Var} \left( \frac{1}{2} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} S_{xy}^{M_i} \left( \frac{\Delta x}{x} \right) \left( \frac{\Delta y}{y} \right) \right) = \mathbb{E} \left( \left( \frac{1}{2} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} S_{xy}^{M_i} \left( \frac{\Delta x}{x} \right) \left( \frac{\Delta y}{y} \right) \right)^2 \right) \quad (43)$$

$$+ \left( \mathbb{E} \left( \frac{1}{2} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} S_{xy}^{M_i} \left( \frac{\Delta x}{x} \right) \left( \frac{\Delta y}{y} \right) \right) \right)^2 \quad (44)$$

Given our independence assumption regarding components, that is  $\mathbb{E} \left( \frac{\Delta x \Delta y}{xy} \right) = 0$ , every cross term cancels.

$$\text{Var} \left( \frac{1}{2} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} S_{xy}^{M_i} \left( \frac{\Delta x}{x} \right) \left( \frac{\Delta y}{y} \right) \right) = \frac{1}{4} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} (S_{xy}^{M_i})^2 \tau_x^2 \tau_y^2 \approx \frac{1}{4} \sum_{x \in \mathcal{P}_i} (S_{xx}^{M_i})^2 \tau_x^4 \quad (45)$$

where our approximation holds when  $S_{xx}^{M_i} \gg S_{xy}^{M_i}$  for all  $x \neq y$ . Suppose for any given stage, the first-order passband variance is some  $V_{\text{target}}$ .

$$g(\tau^2) := -V_{\text{target}} + \sum_{x \in \mathcal{P}_i} (S_x^{M_i})^2 \tau_x^2 = 0 \quad (46)$$

where  $\tau^2 = [\tau_x^2]^T \quad \forall x \in \mathcal{P}_i$ . We can make an optimization statement from this to find the maximum second-order deviation.

$$\text{Maximize } f(\tau^2) := \frac{1}{4} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} (S_{xy}^{M_i})^2 \tau_x^2 \tau_y^2 \text{ given the constraint } g(\tau^2) \quad (47)$$

We can then use Lagrange multipliers, where we have the following Lagrangian.

$$\mathcal{L}(\tau^2, \lambda) = \frac{1}{4} \sum_{x \in \mathcal{P}_i} \sum_{y \in \mathcal{P}_i} (S_{xy}^{M_i})^2 \tau_x^2 \tau_y^2 - \lambda \left( -V_{\text{target}} + \sum_{x \in \mathcal{P}_i} (S_x^{M_i})^2 \tau_x^2 \right) \quad (48)$$

Suppose  $k \in \mathcal{P}_i$ . Then, we differentiate with respect to  $\tau_k^2$ .

$$\frac{\partial \mathcal{L}}{\partial \tau_k^2} = \frac{1}{2} \sum_{x \in \mathcal{P}_i} (S_{xk}^{M_i})^2 \tau_x^2 - \lambda (S_k^{M_i})^2 \quad (49)$$

We get a minimum or maximum when  $\frac{\partial \mathcal{L}}{\partial \tau_k^2} = 0$ .

$$\frac{1}{2} \sum_{x \in \mathcal{P}_i} (S_{kx}^{M_i})^2 \tau_x^2 = \lambda (S_k^{M_i})^2 \quad \forall k \in \mathcal{P}_i \quad (50)$$

Obviously if  $\lambda = 0$ , we get a minimum. To get an easier closed-form maximum for our  $\Delta\tau^2$ , we'll make the following definitions to relate this equation to standard matrix algebra.

$$H_S^{M_i} := \left[ (S_{xy}^{M_i})^2 \right] \quad \forall x, y \in \mathcal{P}_i \quad (51)$$

$$S_{eq}^{M_i} := \left[ (S_x^{M_i})^2 \right] \quad \forall x \in \mathcal{P}_i \quad (52)$$

That is,  $H_s$  is the squared matrix of second-order sensitivities and  $S_{eq}^{M_i}$  is the vector of squared sensitivities. We can rewrite Eq. (50) like so:

$$\frac{1}{2} H_S^{M_i} \tau^2 = \lambda S_{eq}^{M_i} \quad (53)$$

Supposing  $H_S^{M_i}$  isn't singular at the desired frequency, we get that

$$\tau_{\max}^2 = 2\lambda \left( H_S^{M_i} \right)^{-1} S_{eq}^{M_i} \quad (54)$$

We can then rewrite our original functions.

$$g(\tau^2) = (S_{eq}^{M_i})^T \tau^2 - V_{\text{target}} = 0 \quad f(\tau^2) = \frac{1}{4} (\tau^2)^T H_S \tau^2 \quad (55)$$

We then can solve for  $\lambda$ , with our final variance being  $f(\tau_{\max}^2)$ .

$$\lambda = \frac{V_{\text{target}}}{2 \left( (S_{eq}^{M_i})^T \left( H_S^{M_i} \right)^{-1} S_{eq}^{M_i} \right)} \quad (56)$$

We can  $S_{eq}^{M_i}, H_S^{M_i}$  compute analytically, but the expressions are complex enough that it's probably best to compute this numerically at the desired frequency.

It's important to note that if  $K \neq 1$ , we get an asymptote in  $Q$ , which can balloon error to the point that high  $Q$  stages won't be bounded generally by bounding the Taylor expansion.

## VII. LOWPASS FILTER DESIGN

With all of these theoretical results being said, we now will interpret these results to make filter design recommendations.

Recalling Eq. (22), for low  $\rho_i$ , we see a blowup of error due to a vertical asymptote when  $K \neq 1$ , with a similar growth rate at high  $\rho_i$  similar to  $K = 1$ . This implies that  $K = 1$  is typically optimal for minimizing variation, as shown in Figure 2. Though, for low  $Q_i$ , we can use  $K \neq 1$  architectures as the error is already small.

We'll assume for our optimal filter that  $R_{1,i} = mR_{2,i}$ ,  $C_{1,i} = nC_{2,i}$ , and  $K = 1$ , which, as stated by [6]:

$$\omega_i = \frac{1}{R_{2,i}C_{2,i}\sqrt{mn}} \quad Q_i = \frac{\sqrt{mn}}{m+1} \quad (57)$$

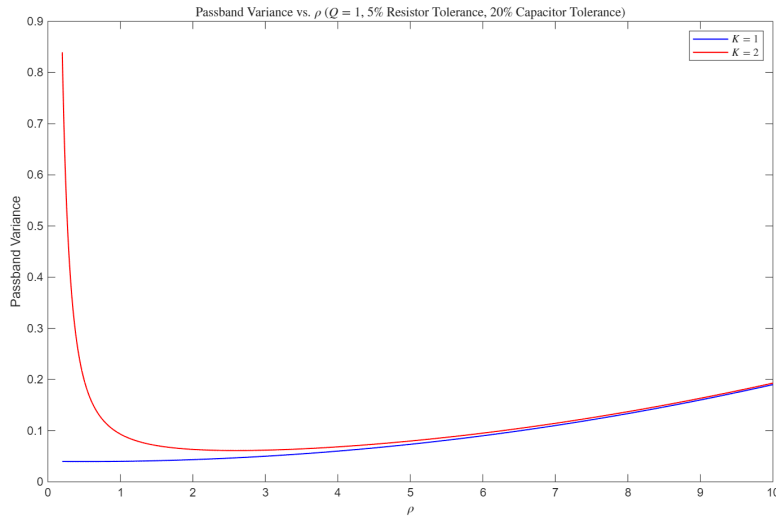


Fig. 2: Our variance bound with respect to  $\rho_i$  for a Sallen-Key stage of  $Q_i = 1$ . Note the asymptote when  $K \neq 1$ .

### A. Designing the Optimal Filter

We will force Eq. (25) like so.

$$\rho_i = \frac{1}{\sqrt{mn}} = \frac{1}{2Q_i} \implies \sqrt{mn} = 2Q_i \quad (58)$$

We can then substitute into Eq. (57).

$$Q_i = \frac{2Q_i}{m+1} \implies m = 1 \implies R_{1,i} = R_{2,i} = R \quad \forall Q_i \quad (59)$$

We can then solve for  $n$ .

$$\sqrt{n} = 2Q_i \implies n = 4Q_i^2 \quad (60)$$

We can then relate  $R$  and  $C_{2,i}$ .

$$\omega_i = \frac{1}{RC_{2,i}2Q_i} \implies RC_{2,i} = \frac{1}{2\omega_i Q_i} \quad (61)$$

To design a stage, choose a reasonable  $R$  to make  $C_{2,i}$  also reasonably valued for a given corner frequency. These two values, coupled with our target  $Q_i$ , give us our  $n$  to find  $C_{1,i}$ . “Reasonable” here also implies we want to get as close to E24 as possible.

### B. Designing a Practical Filter

Though the previous design strategy works very well, we often run into nonstandard capacitor values that may be inconvenient to filter design.

So, we need a similar method that can provide more standard capacitor values. For this new method, we will first choose  $n$  and  $C_{2,i}$ .

$$\frac{Q_i}{\sqrt{n}} = \frac{\sqrt{m}}{m+1} \quad (62)$$

Let  $u := \sqrt{m}$  and  $v = \frac{Q_i}{\sqrt{n}}$ , then,

$$vu^2 - u + v = 0 \implies u = \frac{1 + \sqrt{1 - 4v^2}}{2v} \implies m = \left( \frac{1 + \sqrt{1 - \frac{4Q_i^2}{n}}}{\frac{2Q_i}{\sqrt{n}}} \right)^2 \quad (63)$$

We can then let  $\rho_i = \frac{1}{\sqrt{mn}}$ . Our goal is to get  $\rho_i$  as close to  $\frac{1}{2Q_i}$  as possible while using standard component sizes. Note that we can solve for  $R_{2,i}$  as constrained by the corner frequency.

$$R_{2,i} = \frac{\rho_i}{\omega_i C_{2,i}} \quad (64)$$

### C. Highpass Filter Design

For highpass filter design, make similar assumptions as before (i.e.  $R_{1,i} = mR_{2,i}$ ,  $C_{1,i} = nC_{2,i}$ , and  $K = 1$ ) and perform the same analysis as above, swapping resistors and capacitors. For example, when in the lowpass method we choose  $R$  or  $n$ , we choose  $C$  and  $m$  in the highpass method.

## VIII. SIMULATIONS TO CONFIRM RESULTS

To ensure that the bounds we have are sensible and not too pessimistic, we conduct monte-carlo simulations on three different designs of an 8th-order Butterworth filter with a corner frequency  $f_c = 1\text{kHz}$ . For the Butterworth, the  $Q$  factors are  $Q = 0.5098, 0.6013, 0.9, 2.5628$  respectively. We assume a 10% resistor tolerance and a 20% capacitor tolerance to give a worst-case scenario to our bounds. If we assume that within these tolerance,  $\Delta R, \Delta C$  are approximately normally distributed around 0 and follow that  $\pm 3\sigma_{\Delta R} \approx \pm 10\%R, \pm 3\sigma_{\Delta C} \approx \pm 20\%C$ , then the variances ( $\tau_R^2, \tau_C^2$ ) are the following:

$$\tau_R^2 = \frac{\sigma_R^2}{R^2} = \frac{(10\%)^2}{3} \quad \tau_C^2 = \frac{\sigma_C^2}{C^2} = \frac{(20\%)^2}{3} \quad (65)$$

We can then use Eq. (22) to calculate the variance of each stage. From there, we can convert to dB, considering that if  $|H_i(j\omega)|$  drifts by some  $|H_i(j\omega)|(1 + \frac{\sigma_{|H_i(j\omega)|}}{|H_i(j\omega)|})$ , where the second term is the square root of polynomial value, then

$$\sigma_{dB} = 20 \log_{10}(1 + \sqrt{\text{bound at some } \rho}) \quad (66)$$

Then, with 99% confidence (assuming the normal distribution of components leads to a normal distribution in transfer function magnitude), our maximum passband ripple is  $\pm 3\sigma_{dB}$ .

### A. Optimal Eighth Order Butterworth

Figure 3 shows the simulation setup in LTSpice. We're using an ideal, single-pole opamp in these simulations, and we run the random trial  $n = 500$  times, which should be a large enough sample size to allow the law of large numbers to hold. We predict the maximum transfer function variance to be  $\approx \pm 9.4185\text{dB}$  from our variance bound, which we used a voltage divider and a noninverting opamp configuration to show on the LTSpice graph.

Figures 4 and 5 provide the output of our AC analysis. As we can see, a large majority of the simulations fall under our bounds (the small few that leak come from the  $\pm 3\sigma_{dB}$  approximation of the max ripple).

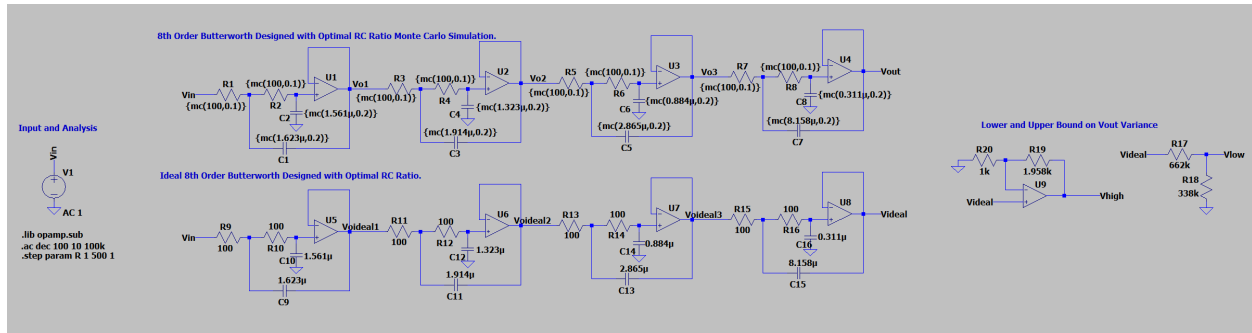


Fig. 3: The LTSpice setup of our optimal eighth-order Butterworth.

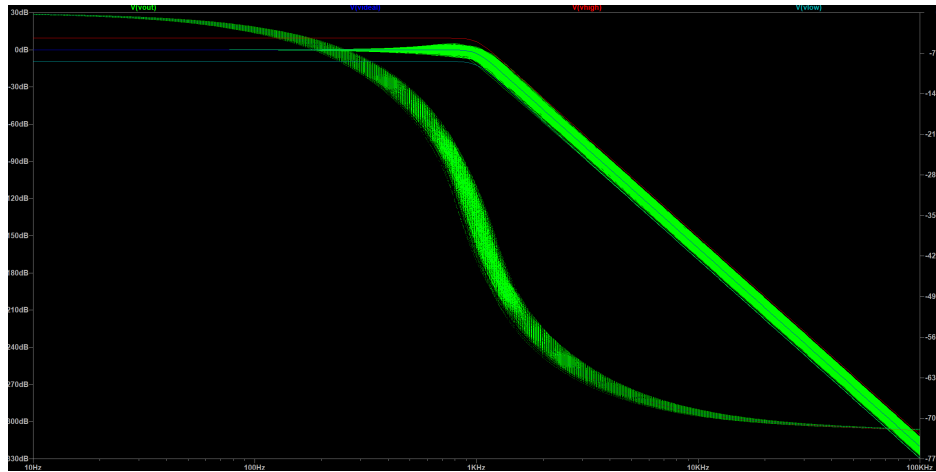


Fig. 4: The results of our optimal eighth-order Butterworth Monte Carlo simulation (wide view).

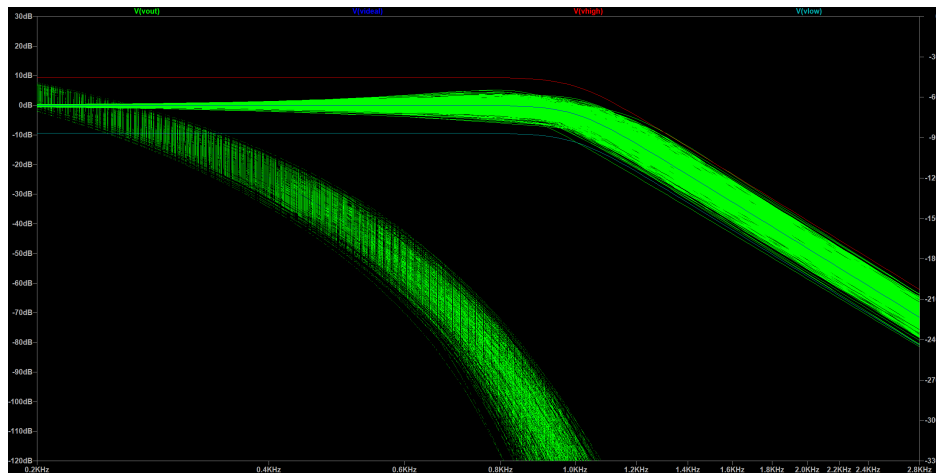


Fig. 5: The results of our optimal eighth-order Butterworth Monte Carlo simulation (zoomed in on corner frequency).

### B. Equal RC Filter (where Taylor approximation fails)

We also run the same process for an eighth order filter where  $C_{1,i} = C_{2,i} = 100$  nF and  $R_{1,i} = R_{2,i} = 1.6$  k $\Omega$ , with our  $Q$  factor governed by the following

$$Q = \frac{1}{3 - K} \quad (67)$$

Figure 6 shows our LTSpice setup, with our results shown in Figures 7 and 8. We can see that our bound for max ripple is  $\approx \pm 16.8643$  dB, which is significantly higher than the  $K = 1$  case. We see many trials breaking this bound with a spike at the corner frequency, with a drastically different phase shift (settling at  $-360^\circ$  instead of  $-720^\circ$ ). The cause of these anomalies is the mismatching tolerance causing  $Q$  to push towards (and possibly cross) an asymptote. To illustrate this phenomenon, consider the last stage of the filter  $Q = 2.5628 \Rightarrow K \approx 2.61$ . If we assume the lowest resistor ratio  $m$  and the highest capacitor ratio  $n$ . By [6],

$$Q = \frac{\sqrt{mn}}{m + 1 + mn(1 - K)} \quad (68)$$

Our Sallen-Key becomes unstable when  $m + 1 \leq mn(K - 1)$ . Then, there are values within our tolerance (i.e.  $\min(m) = \frac{0.9}{1.1}$ ,  $\max(n) = \frac{1.2}{0.8}$ ) which make our Sallen-Key stage unstable. Thus, for any high  $Q$  in the architecture (like the last stage of our filter), our variance bound will break down due to asymptotic increase or the circuit becoming unstable.

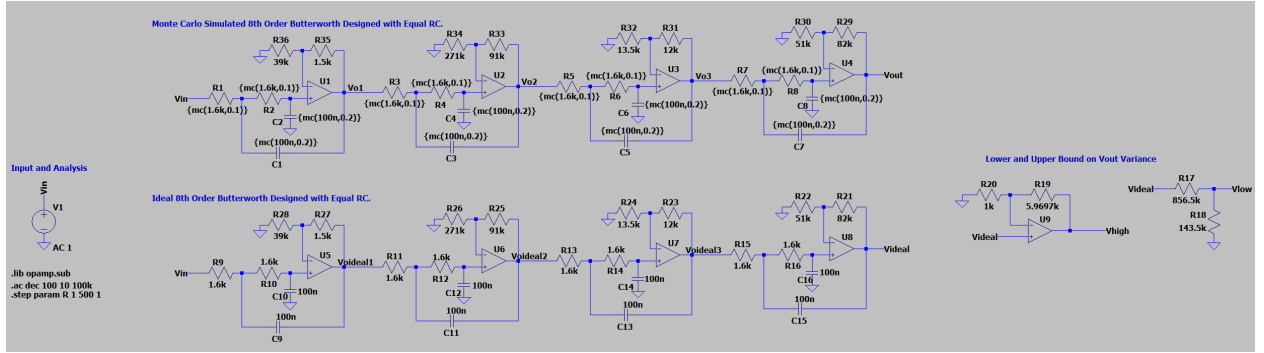


Fig. 6: The LTSpice setup of our equal RC eighth order Butterworth.

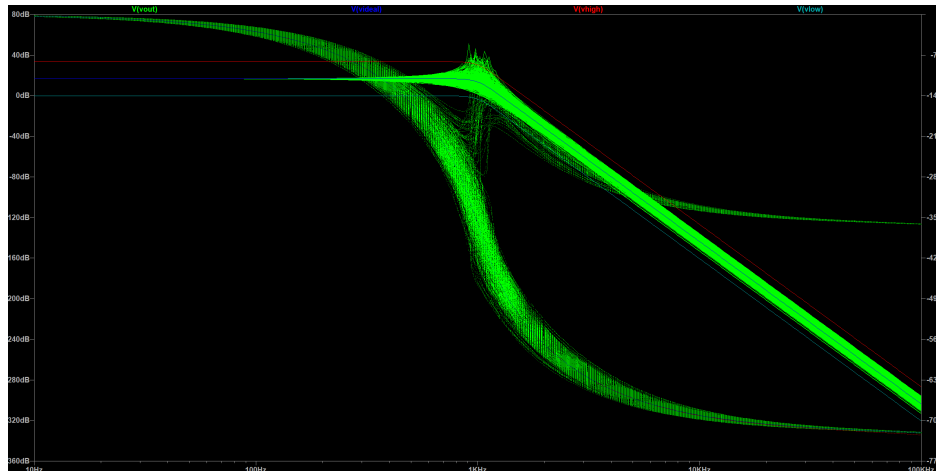


Fig. 7: The results of our equal RC eighth-order Butterworth Monte Carlo simulation (wide view).

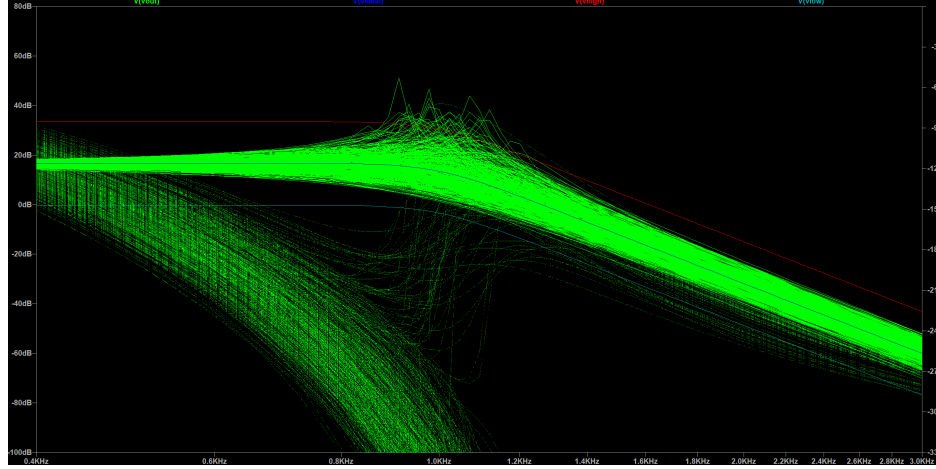


Fig. 8: The results of our equal RC eighth-order Butterworth Monte Carlo simulation (zoomed in on corner frequency).

### C. Practical Eighth Order Butterworth

Finally, we follow the recommendations of Section VII-B. We design a filter that uses standard capacitor values and resistor values up to the point of putting two standard resistor values in series. We consider the values in Figure 12 to be standard for this exercise. Figure 9 shows the filter setup, and Figures 10 and 11 present our results. Our max ripple bound is  $\approx \pm 9.496\text{dB}$ , which is only  $0.0774\text{dB}$  ( $\approx 0.9\%$ ) looser than our optimal filter. One outlier still exists because of our  $\pm 3\sigma_{dB}$  approximation.

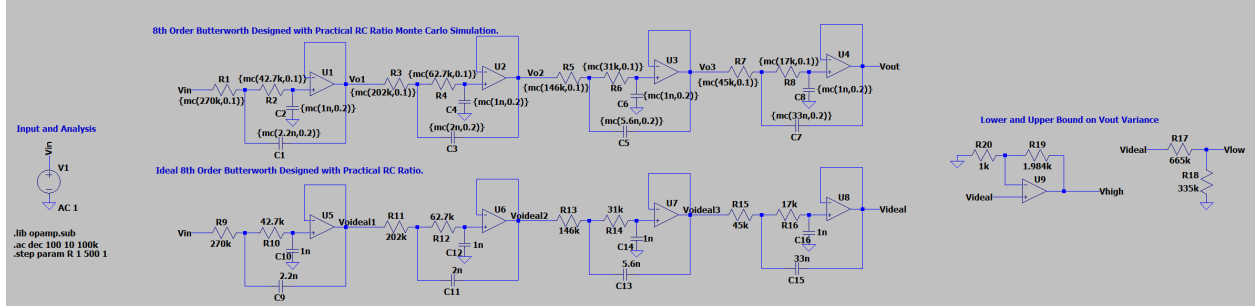


Fig. 9: The LTSpice setup of our practical eighth order Butterworth.

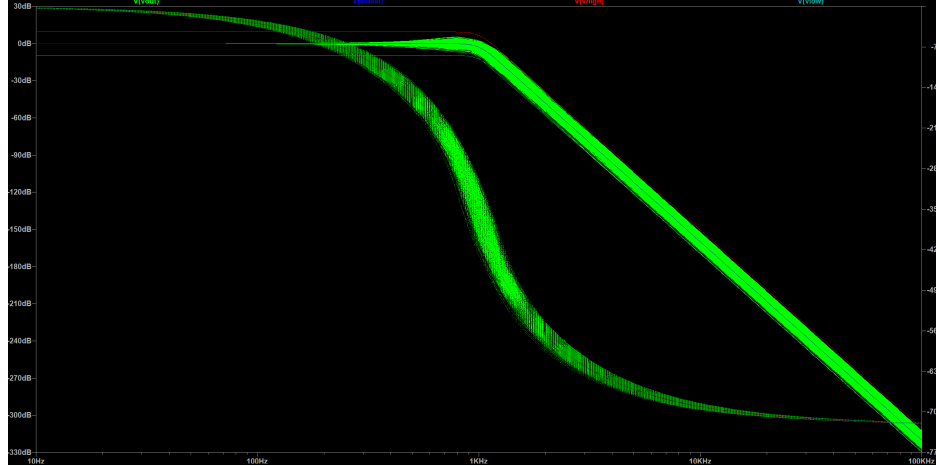


Fig. 10: The results of our practical eighth-order Butterworth Monte Carlo simulation (wide view).

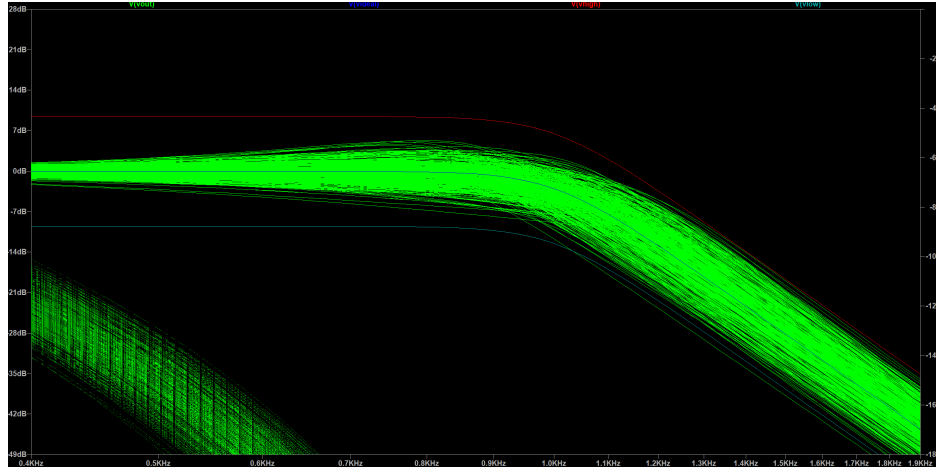


Fig. 11: The results of our practical eighth-order Butterworth Monte Carlo simulation (zoomed in on corner frequency).

## IX. CONCLUSION AND FUTURE WORK

We derived closed-form expressions for the first and second order effects of variance for filters made of Sallen-Key stages. These equations gave us insights as to how Sallen-Keys respond to typical variations of important parameters. We've proven formally that any passband gain over  $K = 1$  is suboptimal for low-variance filter design, with the optimal design being when our resistors or capacitors match in value. From these insights, we've made design recommendations for minimum-variance Sallen-Key stages.

Future work can go into many directions. First, reperforming this paper's analysis restricted to the passband (instead of the whole transfer function) may yield a tighter bound that could be more effective for filter design. Also, this analysis should be repeated for standard  $LC$  stages or multifeedback (MFB) stages. Since MFB stages are often used for narrowband bandpass filters, variance may be a more important consideration for this architecture. More universal and clever application of bounding results from sensitivity analysis can lead to filter designs more resilient to parameter variation.

# APPENDIX A SUPPLEMENTARY FIGURES

| Picofarad | Nanofarad | Microfarad | Code | Picofarad | Nanofarad | Microfarad | Code |
|-----------|-----------|------------|------|-----------|-----------|------------|------|
| 10        | 0.01      | 0.00001    | 100  | 4700      | 4.7       | 0.0047     | 472  |
| 15        | 0.015     | 0.000015   | 150  | 5000      | 5.0       | 0.005      | 502  |
| 22        | 0.022     | 0.000022   | 220  | 5600      | 5.6       | 0.0056     | 562  |
| 33        | 0.033     | 0.000033   | 330  | 6800      | 6.8       | 0.0068     | 682  |
| 47        | 0.047     | 0.000047   | 470  | 10000     | 10        | 0.01       | 103  |
| 100       | 0.1       | 0.0001     | 101  | 15000     | 15        | 0.015      | 153  |
| 120       | 0.12      | 0.00012    | 121  | 22000     | 22        | 0.022      | 223  |
| 130       | 0.13      | 0.00013    | 131  | 33000     | 33        | 0.033      | 333  |
| 150       | 0.15      | 0.00015    | 151  | 47000     | 47        | 0.047      | 473  |
| 180       | 0.18      | 0.00018    | 181  | 68000     | 68        | 0.068      | 683  |
| 220       | 0.22      | 0.00022    | 221  | 100000    | 100       | 0.1        | 104  |
| 330       | 0.33      | 0.00033    | 331  | 150000    | 150       | 0.15       | 154  |
| 470       | 0.47      | 0.00047    | 471  | 200000    | 200       | 0.2        | 204  |
| 560       | 0.56      | 0.00056    | 561  | 220000    | 220       | 0.22       | 224  |
| 680       | 0.68      | 0.00068    | 681  | 330000    | 330       | 0.33       | 334  |
| 750       | 0.75      | 0.00075    | 751  | 470000    | 470       | 0.47       | 474  |
| 820       | 0.82      | 0.00082    | 821  | 680000    | 680       | 0.68       | 684  |
| 1000      | 1.0       | 0.001      | 102  | 1000000   | 1000      | 1.0        | 105  |
| 1500      | 1.5       | 0.0015     | 152  | 1500000   | 1500      | 1.5        | 155  |
| 2000      | 2.0       | 0.002      | 202  | 2000000   | 2000      | 2.0        | 205  |
| 2200      | 2.2       | 0.0022     | 222  | 2200000   | 2200      | 2.2        | 225  |
| 3300      | 3.3       | 0.0033     | 332  | 3300000   | 3300      | 3.3        | 335  |

|     |     |       |       |       |       |
|-----|-----|-------|-------|-------|-------|
| 3.9 | 43  | 470   | 5.1 k | 56 k  | 620 k |
| 4.3 | 47  | 510   | 5.6 k | 62 k  | 680 k |
| 4.7 | 51  | 560   | 6.2 k | 68 k  | 750 k |
| 5.1 | 56  | 620   | 6.8 k | 75 k  | 820 k |
| 5.6 | 62  | 680   | 7.5 k | 82 k  | 910 k |
| 6.2 | 68  | 750   | 8.2 k | 91 k  | 1.0 M |
| 6.8 | 75  | 820   | 9.1 k | 100 k | 1.2 M |
| 7.5 | 82  | 910   | 10 k  | 110 k | 1.5 M |
| 8.2 | 91  | 1.0 k | 11 k  | 120 k | 1.8 M |
| 9.1 | 100 | 1.1 k | 12 k  | 130 k | 2.2 M |
| 10  | 110 | 1.2 k | 13 k  | 150 k | 2.4 M |
| 11  | 120 | 1.3 k | 15 k  | 160 k | 2.7 M |
| 12  | 130 | 1.5 k | 16 k  | 180 k | 3.3 M |
| 13  | 150 | 1.6 k | 18 k  | 200 k | 3.6 M |
| 15  | 160 | 1.8 k | 20 k  | 220 k | 3.9 M |
| 16  | 180 | 2.0 k | 22 k  | 240 k | 4.7 M |
| 18  | 200 | 2.2 k | 24 k  | 270 k | 5.6 M |
| 20  | 220 | 2.4 k | 27 k  | 300 k | 6.8 M |
| 22  | 240 | 2.7 k | 30 k  | 330 k | 8.2 M |

Fig. 12: Standard Resistor/Capacitor Values

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